

Demand-Based Asset Pricing:

A Note on the Framework, Non-Homotheticity, and Negative (Net) Portfolio Shares

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1 Introduction

Demand-based asset pricing reframes the canonical asset-pricing question. Rather than deriving prices from a representative-agent stochastic discount factor and asking which preferences and shocks rationalise the cross-section, it specifies an estimable system of investor-level demand functions over assets, identifies its parameters from observed holdings, and lets prices clear the resulting demand against fixed supply. The intellectual lineage runs from McFadden’s discrete-choice demand (McFadden, 1974), to Berry’s inversion approach (Berry, 1994) and the BLP random-coefficients framework (Berry et al., 1995; Nevo, 2001), to Kojien and Yogo (2019) who first ported these tools to portfolio choice. The framework has since underpinned work on inelastic markets (Gabaix and Kojien, 2021), central-bank asset purchases (Kojien et al., 2021), exchange rates (Kojien and Yogo, 2020), benchmark-driven flows (Pavlova and Sikorskaya, 2023), and the allocation of price-setting power across investor types (Kojien et al., 2024; Haddad et al., 2025).

This note has two purposes. First, it lays out the workhorse logit demand system and its identification (Part I). Second, it confronts the framework with two specific concerns that arise when one tries to use it to model *household* balance sheets across housing, deposits, and equities as income grows (Part II): (i) whether the framework accommodates *non-homotheticity*—portfolio shares that vary systematically with wealth or income—and (ii) whether it accommodates *negative net portfolio shares*, of which the leading example is the negative net housing position created by a mortgage. The household case differs from the institutional case in a way that matters for question (ii): the negative leg (the mortgage) is bundled with the positive leg (the house) by collateral and by a common price, so the “shorts as supply” fix that works for independent short books does not transplant.

PART I: INSTITUTIONAL DEMAND-BASED ASSET PRICING

2 The Logit Demand System for Portfolios

Fix a date t and let i index investors and $n = 1, \dots, N_t$ index inside assets, with $n = 0$ denoting an outside asset (cash, or, more generally, the residual claim on non-asset wealth). Each investor has assets under management $A_{i,t}$ and chooses portfolio weights $\{w_{i,t}(n)\}_{n=0}^{N_t}$ summing to one. Kojien and Yogo (2019) specify the workhorse *characteristics-based logit*:

$$w_{i,t}(n) = \frac{\delta_{i,t}(n)}{1 + \sum_{m=1}^{N_t} \delta_{i,t}(m)}, \quad n = 1, \dots, N_t, \quad (1)$$

$$w_{i,t}(0) = \frac{1}{1 + \sum_{m=1}^{N_t} \delta_{i,t}(m)}, \quad (2)$$

$$\delta_{i,t}(n) = \exp\left(\beta_{i,t}^0 m_{e,n,t} + \beta_{i,t}^\top x_{n,t} + \varepsilon_{i,t}(n)\right), \quad (3)$$

where $me_{n,t}$ is log market equity (or another scale-of-the-asset variable), $x_{n,t}$ is a vector of observable asset characteristics (profitability, dividends-to-book, betas, etc.), and $\varepsilon_{i,t}(n)$ is the latent demand for asset n by investor i .

Three structural features of (1) matter for what follows. First, since $\delta_{i,t}(n) > 0$, every share $w_{i,t}(n) \in (0, 1)$: *negative weights are ruled out by construction*. Second, the system is Cobb-Douglas-like in the risky assets: relative weights $w_{i,t}(n)/w_{i,t}(m)$ depend only on relative δ 's, so they are *scale-free* in $A_{i,t}$. Third, market clearing is asset by asset,

$$p_{n,t} s_{n,t} = \sum_i A_{i,t} w_{i,t}(n), \quad (4)$$

where $s_{n,t}$ is shares outstanding, so the equilibrium price $p_{n,t}$ is implicitly defined and can be solved fixed-point style. The map from demand parameters to equilibrium prices is what makes counterfactuals (Was QE effective? Are markets inelastic? Who set the price of Tesla?) tractable.

3 Identification and Estimation

The empirical challenge in (3) is that $me_{n,t}$ and the latent demand $\varepsilon_{i,t}(n)$ are both functions of price and so correlated: assets with high latent demand have high prices and so high market equity. OLS estimates of $\beta_{i,t}^0$ are biased toward zero (or even positive), implying spuriously inelastic demand.

Koijen and Yogo (2019) resolve this with an instrument built from investors' *investment universes*. Let $\mathbf{1}_{i,t}(n) \in \{0, 1\}$ indicate whether asset n is in investor i 's feasible set at t (estimated from the empirical extensive margin of holdings). The instrument for $me_{n,t}$ in investor i 's demand equation is the counterfactual market equity that would obtain if every *other* investor $j \neq i$ held an equal-weighted portfolio over their own universe:

$$\widehat{me}_{n,t}^{(-i)} = \log \left(\sum_{j \neq i} A_{j,t} \frac{\mathbf{1}_{j,t}(n)}{\sum_m \mathbf{1}_{j,t}(m)} \right). \quad (5)$$

The exclusion restriction is that other investors' AUM and universe composition shift the supply of capital chasing asset n without shifting i 's latent demand for it. Conditional on asset characteristics, this yields finite (and economically large) own-price elasticities, typically near one—an order of magnitude below what representative-agent calibrations would imply, and the empirical foundation of the inelastic-markets literature (Gabaix and Koijen, 2021).

The instrument has been the subject of two important refinements. van der Beck (2022) shows that when latent demand has a common factor structure, the instrument is partly endogenous and elasticities are biased; he proposes corrections that anchor the estimated demand system to a small set of fundamentals. Haddad et al. (2025) and Koijen et al. (2024) examine heterogeneity in elasticities across investor types and find that a small set of large, active investors do most of the price setting.

4 Non-Homotheticity in the Institutional Setting

The bare logit (1) is homothetic in $A_{i,t}$ *within the risky set*: doubling AUM holding the δ 's fixed scales every position proportionally and leaves every share unchanged. To deliver shares that move systematically with size, the framework must encode size in either the coefficients or the characteristics. Four common hooks:

1. Let $\beta_{i,t}^0$ depend on $\log A_{i,t}$, so the loading on log market equity is size-elastic. Equivalently, allow the elasticity of demand to vary with investor size.

2. Interact asset characteristics with investor characteristics: $\beta_{i,t} = B \cdot z_{i,t}$ for a matrix B and a vector $z_{i,t}$ of investor attributes. [Kojien et al. \(2024\)](#) and [Kojien and Yogo \(2020\)](#) pursue this aggressively, with separate β 's estimated by investor type.
3. Allow the outside-asset margin $w_{i,t}(0)$ to depend on AUM through a size-dependent constant in (3).
4. Use random-coefficient (BLP-style) extensions in which the distribution of $\beta_{i,t}$ over investors has wealth-varying moments ([Berry et al., 1995](#); [Nevo, 2001](#)).

All four are parametric: non-homotheticity is imposed through functional form on the coefficients, not derived from preferences. The distinction matters when extending to households (Section 7 below), where preference-based theories of non-homotheticity already exist ([Wachter and Yogo, 2010](#)).

5 Independent Short Selling

Because $\delta_{i,t}(n) > 0$, the logit cannot produce a negative $w_{i,t}(n)$. In the original [Kojien and Yogo \(2019\)](#) application this is benign—the data are 13F long-only holdings, where shorts are mostly absent. Four routes accommodate shorts when they matter:

1. *Augmented choice set*: treat each asset as two products, a long position and a short position, each with its own non-negative logit share; net exposure is long minus short.
2. *“Shorts as supply” (twin logit)*: the investor brings a long-side demand with budget $A_{i,t}^L$ and a short-side supply with budget $A_{i,t}^S$, the latter bounded by the securities-lending market ([D’Avolio, 2002](#); [Duffie et al., 2002](#); [Vayanos and Weill, 2008](#); [Lewellen, 2011](#)). Market clearing becomes net, and signed exposure emerges in equilibrium.
3. *Mean-variance or linear demand*: naturally signed, at the cost of losing the discrete-choice microfoundation and characteristics parsimony.
4. *Local linearisation around a no-short benchmark*: elasticities are recovered analytically while estimation is restricted to long holdings ([Haddad et al., 2025](#)).

Crucially, each of these treats the short as *independent* of any particular long position: the investor freely allocates shorting capacity across assets, subject to a separate budget. This is the appropriate stylisation for an equity hedge fund, and it is not the right stylisation for a mortgaged homeowner. Part II addresses why.

PART II: HOUSEHOLD DEMAND SYSTEMS WITH LEVERAGE AND NON-HOMOTHETICITY

6 From Institutional to Household Demand Systems

Transposing the logit framework to households changes several primitives. The choice set narrows from thousands of securities to a handful of asset *classes*: housing, deposits, listed equities, bonds, and private business equity. The outside-asset margin, a residual in the institutional case, becomes a first-order modelling object—housing dominates the median household balance sheet. Income, not just wealth, is a structural driver of allocations, so the investor-characteristic vector $z_{i,t}$ must include labour-income variables. Leverage is no longer optional; many household balance sheets are built around a mortgage. Finally, the relevant data are surveys (the U.S. Survey of Consumer Finances, the euro-area Household Finance and Consumption Survey) or administrative wealth registers (Sweden, Norway), not 13F.

The demand-system literature on households is younger than its institutional counterpart but already extends to insurance choice ([Kojien et al., 2016](#)) and is being applied to balance-sheet

composition in ongoing work. Its appeal is the same as in the institutional setting: characteristics-based parsimony, an explicit demand function rather than a structural utility, and the prospect of identifying elasticities from cross-sectional variation rather than from contested time-series moments.

7 Non-Homothetic Household Portfolio Shares

The household-finance literature documents that portfolio shares are strongly non-homothetic. The risky share rises with wealth, both within and across cohorts (Calvet et al., 2007, 2009; Carroll, 2002; Bach et al., 2020; Fagereng et al., 2020); gross housing’s share of total assets declines with wealth above a threshold, while deposits are hump-shaped or declining; mortgage debt is hump-shaped over the life cycle. The Gomes et al. (2021) survey collects this evidence.

Theoretically, two routes generate this empirically. The *structural* route derives non-homotheticity from preferences and constraints: HARA utility, subsistence consumption, luxury goods, or borrowing constraints that bind differently at different wealths (Wachter and Yogo, 2010; Cocco, 2005; Yao and Zhang, 2005; Chetty et al., 2017). The *demand-system* route, in contrast, imposes non-homotheticity through the parametric hooks listed in Section 4: let $\beta_{i,t}$ in (3) depend on $(\log A_{i,t}, \log y_{i,t}, \text{age}_{i,t})$ so that the loading on each asset characteristic moves with wealth, income, and age. Concretely, for asset class $n \in \{H, D, E\}$ (housing, deposits, equities),

$$\delta_{i,t}(n) = \exp\left(\alpha_n + \beta_n^w \log A_{i,t} + \beta_n^y \log y_{i,t} + \beta_n^a \text{age}_{i,t} + \gamma_n^\top x_{n,t} + \varepsilon_{i,t}(n)\right), \quad (6)$$

where the wealth-, income-, and age-dependent intercepts deliver non-homothetic shares directly. Coefficients β_n^w, β_n^y are identified from the cross-section of households with different (A, y, age) holding the same characteristics $x_{n,t}$ fixed.

This route is flexible but *descriptive*: (6) can reproduce any cross-sectional pattern of shares in (A, y, age) but does not explain why. For applications where the question is “what would happen to allocations if income y moved by Δ ,” this is enough; for welfare or policy counterfactuals, one usually wants a structural foundation alongside.

8 Linked Positions: Housing and Mortgages

The user’s motivating case—negative net housing shares created by mortgages—fails the institutional fixes of Section 5 for one reason. A mortgage is not an independent short. It is the financing leg of a specific housing purchase, tied to the corresponding gross housing position by a loan-to-value constraint

$$M_{i,t} \leq \ell_{i,t} H_{i,t}, \quad (7)$$

and by a common underlying price (the house price). Modelling a “demand for mortgages” as an independent supply, separately characterised and budgeted, severs that link. Four reformulations preserve it.

Gross-portfolio reformulation with explicit LTV. The cleanest route, and the one we recommend for the user’s application. Run the logit (1)–(6) over *gross*, non-negative positions $\{H, D, E, B, M\}$, treating the mortgage M as a separate “asset” whose characteristic vector includes a negative-yield component (the mortgage rate) and whose choice is linked to H via (7). Net portfolio shares are then accounting identities:

$$\frac{H_{i,t} - M_{i,t}}{A_{i,t}}, \quad \frac{D_{i,t}}{A_{i,t}}, \quad \frac{E_{i,t}}{A_{i,t}}, \quad \dots, \quad (8)$$

with $A_{i,t} = H_{i,t} + D_{i,t} + E_{i,t} + B_{i,t} - M_{i,t}$, and they can take any sign. Negative net housing shares arise mechanically for young leveraged households where $M_{i,t} > H_{i,t}$ (underwater) or

where $H_{i,t}$ enters $A_{i,t}$ in the denominator only through $H - M$. Non-homothetic *net* housing shares follow from non-homothetic *gross* H and M schedules in (6), which scale differently in (A, y, age) .

The cost of this approach is that (7) is an inequality constraint, so closed-form market clearing breaks when it binds—one solves the household problem subject to the constraint and aggregates numerically. Identification of the elasticity of M to the mortgage rate exploits monetary-policy shocks and lender-market structure (Greenwald, 2018; Kaplan et al., 2020).

Nested logit with continuous leverage. An alternative when leverage choice is continuous: an outer logit over asset-class shares (Section 7), then an inner Tobit / fractional-response model for $\ell_{i,t} \in [0, \bar{\ell}]$ conditional on $H_{i,t} > 0$. This keeps the IO discrete-choice machinery for the “what to hold” decision and isolates the leverage choice for separate identification (e.g., from regional variation in lender competition).

Bundle-as-product. If leverage is effectively discrete (a handful of mortgage products), treat “housing at LTV ℓ_k ” as the k -th product entering the logit alongside deposits and equities. Non-negativity is preserved at the bundle level, and net housing share is mechanically $(1 - \ell_k) w_{H, \ell_k}$. The choice set grows, but the framework is unchanged.

AIDS / QAIDS on net shares. The most direct way to allow signed and non-homothetic shares is to abandon logit for an Almost Ideal Demand System (Deaton and Muellbauer, 1980) or its quadratic extension (Banks et al., 1997). These specify shares as linear (or quadratic) functions of log-prices and log-total-expenditure, so they are naturally signed and the “Q” term encodes non-homotheticity directly. The trade-off is the loss of three things the logit gave us: the McFadden microfoundation, characteristics-based parsimony (AIDS is specified at the product level, not the characteristic level), and the Koijen and Yogo (2019) IV strategy. For exploratory empirical work on household balance sheets, AIDS/QAIDS may in fact be the better starting point.

9 Mapping to Household Data

The gross-portfolio reformulation aligns naturally with how household data are organised. The SCF (US), the HFCS (euro area), and Swedish or Norwegian administrative wealth registers all report assets gross of liabilities (gross housing value, deposit balances, mutual-fund holdings, business equity) and liabilities as separate line items (primary-residence mortgage, other property loans, consumer credit, student loans). The implementation maps directly:

- Gross H, D, E, B, M from the wealth statement.
- Investor characteristics $z_{i,t}$ from the demographic and income blocks: age, education, family structure, labour income, unemployment risk proxies.
- Asset characteristics $x_{n,t}$: deposit and mortgage rates at the local lender level, regional house-price-to-rent ratios, expected equity-market characteristics (dividend yield, P/E) at the household’s investment horizon.

Identification is harder than in the institutional case. The Koijen and Yogo (2019) instrument relies on rich heterogeneity in investment universes across investors; households’ universes are narrow and overlap heavily. Substitutes used in the household-finance literature include: regional house-price shocks (Kaplan et al., 2020); monetary-policy-driven mortgage-rate variation (Greenwald, 2018); lender market structure; cohort and age effects for life-cycle non-homotheticity; idiosyncratic income shocks. The natural empirical strategy is to combine characteristics-based demand estimation à la Koijen-Yogo with quasi-experimental identification of a small number of key elasticities, in particular the mortgage-rate elasticity of M conditional on H .

Pitfalls worth flagging: top-coding of high wealth in surveys, missingness in private business equity, mark-to-market versus historical-cost housing values, mortgage-refinancing timing that breaks contemporaneous identification of M from current rates, and the fact that LTV constraints (7) bind selectively across the wealth distribution.

10 Concluding Remarks

The logit demand framework of Koijen and Yogo (2019) can be transposed to household balance sheets but requires two extensions to deliver what the user’s application needs. Non-homotheticity (the allocation pattern of housing, deposits, and equities as income grows) is accommodated parametrically via wealth-, income-, and age-interacted coefficients in (6)—flexible but descriptive. Negative *net* portfolio shares (the negative net housing position created by a mortgage) are accommodated naturally by a gross-portfolio reformulation with an explicit LTV constraint, where gross positions remain non-negative inside the logit and net shares are accounting identities that can take any sign. The independent-short fixes that work for institutional equity portfolios—augmented choice set or “shorts as supply”—do *not* fit the household-leverage case because the negative liability is bundled with a specific positive asset.

For analysts willing to give up the discrete-choice microfoundation and the Koijen-Yogo IV, an AIDS or QAIDS system on net shares is the closer-fitting tool: it is naturally signed, naturally non-homothetic, and has decades of empirical infrastructure behind it. The decision between these two routes turns on whether the priority is characteristic-based counterfactual analysis (favour the gross-portfolio logit) or flexible description of the wealth-allocation profile (favour QAIDS).

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